Maharashtra State Board Of Technical Education, Mumbai

Learning and Assessment Scheme Of Diploma Courses for Working Professionals

Programme Name	: Diploma In Mining & Mine Surveying (Working Professional)		
Programme Code	: PMS	**With Effect From Academic Year**	: 2024-25
Duration of Programme	: 3 Year		
Year	: Second	**Scheme**	: F

Sr No	Course Title	Abbrevation	Course Type	Course Code	Diploma Awarding Course	Total IKS Hrs for Sem.	Learning Scheme: Actual Contact Hrs./Week CL	TL	LL	Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week	Credits	Assessment Scheme: Paper Duration (hrs.)	Theory FA-TH Max	SA-TH Max	Total Max	Total Min	Based on LL & TL Practical FA-PR Max	FA-PR Min	SA-PR Max	SA-PR Min	Based on Self Learning SLA Max	SLA Min	Total Marks
(All Compulsory)																								
1	MINE VENTILATION	MVE	DSC	332304	No	0	3	-	2	-	5	5	3	30	70	100	40	25	10	25#	10	-	-	150
2	UNDERGROUND COAL MINING	UCM	DSC	332307	No	1	3	-	2	-	5	5	3	30	70	100	40	25	10	25#	10	-	-	150
3	APPLIED MECHANICS	AMC	AEC	332301	No	1	2	-	2	-	4	4	3	30	70	100	40	25	10	25@	10	-	-	150
4	BASIC SURVEYING	BSU	SEC	332302	No	1	2	-	2	-	4	4	3	30	70	100	40	25	10	25#	10	-	-	150
5	ECONOMIC AND FIELD GEOLOGY	EFG	DSC	332303	No	1	2	-	2	1	5	5	3	30	70	100	40	25	10	25#	10	50	20	200
6	MINING TECHNOLOGY	MTE	DSC	332305	No	1	2	-	2	-	4	4	1.5	30	70*#	100	40	25	10	25#	10	-	-	150
Total						**5**	**14**	**0**	**12**	**1**		**27**		**180**	**420**	**600**		**150**		**150**		**50**		**950**

Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :
1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project./Community (INP) , AbilityEnhancement Course (AEC) , Skill Enhancement Course (SEC) , GenericElective (GE)

Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: MINE VENTILATION**
Course Code	**: 332304**

I. RATIONALE

The underground working is devoid of the natural air. As such to make the working places safe for the persons to work and pass it is necessary to circulate the air artificially through the mine working. A mining engineer should be aware of the principles of how the flow of air can be created, regulated, controlled and monitored. They should be aware of the heat effect and humidity, condition and means of measuring and controlling the same. Number of mine gases is produced in the mine, which has got dangerous and toxic properties.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences: • Evaluate presence of inflammable and toxic/noxious gases in the general body of air and undertake the ventilation survey as per requirement.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use various instruments for detection of inflammable and toxic/noxious gases in the mine.
- CO2 - Measure hunnidity and cooling power of the mine air.
- CO3 - Identify the ventilation system of a mine.
- CO4 - Identify various ventilation appliances used in mine.

- CO5 - Use various instruments in pressure and quantity survey.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332304	MINE VENTILATION	MVE	DSC	3	-	2	-	5	5	3	30	70	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

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4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
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6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the method of detection of given damps. TLO 1.2 Explain the given tests carried out by Flame Safety Lamp. TLO 1.3 Describe Methanometer and measurement of methane using Methanometer. TLO 1.4 Explain principle of Gas Chromatography.	**Unit - I Mine Air** 1.1 Different Gases / Damps found in mines, Definition of damps, their threshold limits, physiological effects, source of production and detection, Degree of gassiness of seam, multi gas detector. 1.2 Flame safety lamps, its principle, construction, safety features, and comparison. Detection of Methane by flame safety lamp. 1.3 Methanometer its principle of working, construction. Principle of following methods of detection of methane. a)Formation of gas cap of Flame safety lamp b)Wheatstone bridge principle c)Diffusion-Combustion - Contraction d)Difference in refractive index of pure air and methane e)Infrared spectrometry f)Reaction of methane with chemicals. 1.4 Principle of chromatography and types of gas chromatography with its applications.	Lecture Using Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on Project based

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain the objectives and standards of ventilation. TLO 2.2 Describe the method of calculation of ventilating pressure. TLO 2.3 Calculate relative humidity of mine air under given condition. TLO 2.4 Interpret effect of heat and humidity on miners. TLO 2.5 Calculate the motive column & Natural Ventilation Pressure (NVP).	**Unit - II Mine Climate** 2.1 Purpose and standards of ventilation, standards for minimum & maximum velocity of air for different locations. 2.2 Measurement of air Pressure, Water gauge, total pressure, Measurement of Ventilating pressure using an inclined Manometer. 2.3 Sources of heat and humidity in mines, Moisture content of mine air, relative humidity, wet bulb temperature, measurement and calculation of relative humidity using Whirling Hygrometer. 2.4 Cooling power of mine air, determination of cooling power by using Kata Thermometer, methods of improving cooling power of mine air, effect of heat and humidity on miners. 2.5 Natural ventilation Pressure, geothermic gradient, Factors causing NVP, Motive column, calculation of natural ventilation pressure.	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on
3	TLO 3.1 Compare given types of fans used for mine ventilation. TLO 3.2 Calculate different efficiencies and theoretical depression produced by fan.	**Unit - III Artificial Ventilation** 3.1 Different types of fans used in mines: centrifugal & axial flow, their principle of working, Exhaust & forcing type. Purposes of evasee & volute casing. Fans in series and parallel, Comparison between axial flow & Centrifugal fan; exhaust & forcing Fan. 3.2 Fan laws, Manometric efficiency overall efficiency, theoretical depression produced by fan. Calculation of water guage, BHP, efficiency, theoretical depression using fan laws.	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Calculate the quantity of air flowing in mines under given condition. TLO 4.2 Describe the given ventilation devices & auxiliary fans. TLO 4.3 Describe the given ventilation devices& auxiliary fans.	**Unit - IV Distribution & Coursing of Air in Mines** 4.1 Laws of air flow in Mines, Atkinson's formula, splitting, its advantages & disadvantages, Calculation of quantity of air in each split, Equivalent orifice: Definition and formula. 4.2 Ventilation appliances, Auxiliary ventilation: Different methods, advantages & disadvantages, hazards associated with auxiliary ventilation, precautions required. Booster fan: purpose, dangers associated, precautions before installation, numerical on booster fan. 4.3 Ascensional and Descensional ventilation, advantages and disadvantages	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on
5	TLO 5.1 Describe given type of ventilation survey. TLO 5.2 Describe given instruments to measure the quantity and pressure of mine air. TLO 5.3 Explain given ventilation survey.	**Unit - V Ventilation Survey** 5.1 Scope and importance of ventilation survey, survey interval and location of survey station, ventilation plan. 5.2 Measurement Of quantity & pressure difference, anemometer, pitot static tube. 5.3 Method of conducting Pressure & quantity survey, precautions during and before conducting ventilation survey, conventional signs used in ventilation survey.	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Detect percentage of Firedamp/Whitedamp/Blackdamp/Stinkdamp/oxygen.	1	*Use of Multigas detector (Drager Xam 5600/Mini Warn)	2	CO1
LLO 2.1 Identify various parts of MSA Methanometer and detecting percentage of methane.	2	*Use of MSA Methanometer.	2	CO1
LLO 3.1 Identify GL5, GL50, GL60 & GL7 flame safety lamps	3	*Flame safety lamps.	2	CO1
LLO 4.1 Dismantle & assemble of GL50 and GL7 Flame safety lamps.	4	*GL50 and GL7 Flame safety lamps.	2	CO1
LLO 5.1 Use flame safety lamp for accumulation test.	5	*Use of Flame safety lamp.	2	CO1
LLO 6.1 Use flame safety lamp for percentage test.	6	*Use of Flame safety lamp.	2	CO1
LLO 7.1 Identify various parts of whirling hygrometer.	7	*Whirling hygrometer.	2	CO2
LLO 8.1 Use whirling hygrometer to determine relative humidity of air.	8	*Use of whirling hygrometer.	2	CO2
LLO 9.1 Use kata thermometer to determine cooling power of mine air	9	*Use of Kata thermometer.	2	CO2
LLO 10.1 Identify various parts of centrifugal fan.	10	*Centrifugal mine fan.	2	CO3
LLO 11.1 Identify various parts of axial flow fan.	11	*Axial flow fan.	2	CO3
LLO 12.1 Identify various parts of air crossing.	12	*Air crossing model.	2	CO4
LLO 13.1 Identify various types of ventilation stopping.	13	*Ventilation stopping model.	2	CO4
LLO 14.1 Simulate working of a regulator.	14	*Regulator model.	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 15.1 Identify various parts of permanent stopping.	15	*Construction of Permanent stopping.	2	CO4
LLO 16.1 Identify various parts of an Air lock	16	Air lock model.	2	CO4
LLO 17.1 Use Vane Anemometer for calculating quantity of air as per given conditions.	17	*Use of Vane Anemometer.	2	CO5
LLO 18.1 Identify various parts of Velometer.	18	*Velometer.	2	CO5
LLO 19.1 Measure air velocity by Velometer.	19	*Use of Velometer	2	CO5
LLO 20.1 Identify various parts of inclined manometer.	20	Inclined manometer	2	CO5
LLO 21.1 Calculate velocity pressure by using Inclined Manometer.	21	Use of Inclined Manometer	2	CO5
LLO 22.1 Identify various parts of Pitot Static Tube.	22	Pitot Static Tube.	2	CO5
LLO 23.1 Calculate static pressure & velocity pressure by using Pitot Static Tube.	23	Use of Pitot Static Tube.	2	CO5
LLO 24.1 Draw various Conventional signs used in Ventilation Plan.	24	Conventional signs used in Ventilation Plan.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Prepare a report on measurement of different gases by using multigas detector by visiting nearby underground mine.
- Prepare a report on mine fan house regarding water gauge by visiting nearby underground mine.
- Prepare report on design parameters of evasee.
- Present a power point presentation on various types of fans used in Indian underground mines for ventilation purpose.
- Prepare any one model of ventilation devices.
- Conduct a ventilation survey in nearby underground mine and prepare a report of it.
- Prepare a report/paper on history of gas testing by old conventional methods.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Multigas detector(drager X-am 5600/ Mini warn:- It has an ergonomic design, and used for detection of exposive, combustible and toxic gases as well as oxygen.	1
2	Model of Centrifugal fan	10
3	Model of Axial flow fan.	11
4	Model of Air crossing.	12
5	Model of Ventilation stopping.	13
6	Model of Regulator.	14
7	Model of Airlock.	16
8	Vane Anemometer :- It is an instrument to determine the distance travelled by air in a given time and is used where the air velocities are between 60m/min and 1000m/min. Its possible measuring wind speed is 1-15m/sec.	17
9	Velometer :- It is an electronic device which directly measure the air velocity in m/sec at any point of observation. Its range varies from 0-15m/sec.	18,19
10	MSA Methanometer :- MSA D6 Methanometer is commonly used in our coal mines, It is suitable for spot checking of gas accumulation with measurement range from 0 to 5 % by volume, The battery lasts for nearly 1000 ten-sec and the complete charging takes about 14 hours.	2
11	Inclined Manometer :- Its is used for measuring pressure/ water gauge in mines. Maximum range of pressure measurement is 250mm Wg.	20,21
12	Pitot Static tube :- Its is used to measure to measure the static pressure, velocity pressure or the total pressure. L-type pitot tube feature a modified ellipsoidal head with a single forward facing hole (for total pressure) and a ring of side holes (for static pressure).	22,23

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
13	Flame safety lamp (GL5, GL50, GL60, GL7) :- It is used for detection of inflammable gases (Firedamp) in underground coal mines, It can detect accumulation and perecntage of firedamp in the atmosphere. It is manufactured by J.K. Dey & Sons, Kolkata.	3,4,5,6
14	Whirling hygrometer :- It is used to determine relative humidity in mine air. Its range generally varies from -5 to 50 degree celcius.	7,8
15	Kata thermometer :- It is used to measure the cooling power of mine air, It consist of an alcohol thermometer with a large bulb 4cm. long and 2cm. in diameter and a stern 20cm. long. It is graduated at two points namely 38 degree celsius and 35 degree celsius	9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Mine Air	CO1	19	4	4	6	14
2	II	Mine Climate	CO2	19	4	4	6	14
3	III	Artificial Ventilation	CO3	20	2	4	10	16
4	IV	Distribution & Coursing of Air in Mines	CO4	18	2	4	10	16
5	V	Ventilation Survey	CO5	14	2	4	4	10
Grand Total				**90**	**14**	**20**	**36**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Progressive test
- Progressive test
- Lab performance
- Journal submission

Summative Assessment (Assessment of Learning)

- End Year Examination
- Viva-voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3			3	3		2			
CO2	3	2		3	3		2			
CO3	3	1	2	3			2			
CO4	3	1	2	3	3		2			

CO5	3	2	2	3	3	1	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	D.J. Deshmukh	Elements of Mining Technology- Vol.2	Dennet, Nagpur, ISBN: 978-81-89904-34-0
2	G.B. Misra	Mine Environment and Ventilation	Oxford University Press, ISBN: 0 19 562232 4
3	L.C. Kaku	Numerical Problems on Mine Ventilation	Lovely Prakashan, Dhanbad, ISBN: 978-8179561393
4	L.C. Kaku	Gas Testing	Lovely Prakashan, Dhanbad, ISBN: 978-8179561522

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://youtu.be/Id4R-2KT9jc?si=xv-Z7yCrcHAewK2n	Use of Whirling Hygrometer to determine relative humidity of mine air.
2	https://youtu.be/tH3O9dlOvfs?si=B__hPrtzFfWqF47h	Types of fans used in mines for artificial ventilation.
3	https://youtu.be/JHb6fkqNyTY?si=Oda0_61_NTa_9cGh	Types of damps.
4	https://youtu.be/N7PSFm680Is?si=5X66qXmZZ9eRNlNW	Types of Flame Safety lamps used in mines.
5	https://youtu.be/Ww5BxnIkj_0?si=kJgOlGKHSC7niqkG	Concept of Water gauge and Ventilation pressure.
6	https://youtu.be/LmctN-Kw75Q?si=MG8upv0CjFl5Q90M	Instruments used in mines for ventilation survey.
7	https://youtu.be/6j6DZGcxocM?si=s59fsBvMGOhzrlsd	Concept of Methanometer for firedamp detection.

Sr.No	Link / Portal	Description

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

UNDERGROUND COAL MINING **Course Code : 332307**

Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: UNDERGROUND COAL MINING**
Course Code	**: 332307**

I. RATIONALE

It is important that the mining diploma students should have knowledge about the common methods of working coal with special reference to Indian Coal Mining. This subject is introduced to understand Methods of Mining of Coal e.g. Board and Pillar working , development and depillaring, Long wall methods both advancing and retreating, special methods for working under special difficult situation and of contiguous seams and thick seam Working. Since new working methods like High wall Mining, Punch Long wall Mining and blasting Gallery method are likely to be practiced in future , these topics have been included this course.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Aim of this course is to help the student to attain the following industry identified outcomes . Select suitable Method of working for underground Coal Mining and Design suitable layouts for Board and Pillar and Long wall Mining.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use appropriate method of working under given geological condition to prepare suitable layout.
- CO2 - Select type of depillaring method for existing roof conditions of the underground coal mines.
- CO3 - Choose the appropriate method of thick seam working and prepare a working plan.
- CO4 - Design long wall working layout under given geological conditions.

- CO5 - Identify the effect of subsidence due to underground working to control .

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332307	UNDERGROUND COAL MINING	UCM	DSC	3	-	2	-	5	5	3	30	70	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 1 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
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V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

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1	TLO 1.1 Classify the given types of method of working. TLO 1.2 Select the suitable method of working. TLO 1.3 Calculate Manpower and OMS under given condition. TLO 1.4 Compare panel System of working on the basis of given criteria.	**Unit - I Board and Pillar Method (Development)** 1.1 Classification of methods of working ,stripping ratio ,geological parameters , technical parameters, history of Indian Coal Mining (IKS). 1.2 Conditions for selection of method of working, Board and pillar applicability advantages and disadvantages, different types development (Strike and Dip development), their advantageous and disadvantages. 1.3 Calculation of percentage of extraction, Design of panel, Different layout classification ,SDL chain conveyor layout ,Continuous miners ,LHD and chain conveyor layout ,Manpower calculation and OMS. 1.4 Open and closed panel system. advantageous and disadvantages ,isolation stopping (Ordinary and explosion proof).	Model Demonstration Video Demonstrations Lecture Using Chalk-Board Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Execute Preparatory arrangements to be done before depillaring. TLO 2.2 Explain types of pillar extraction methods. TLO 2.3 Describe procedure of Goaf Treatment.	**Unit - II Board and Pillar Method (Depillaring)** 2.1 Preparatory arrangements before depillaring, permission from different regulatory authorities, check and joint survey, updatation of plans and sections, construction of isolation and preparatory isolation stopping, Line of extraction and numbering of pillars, Systematic support rules. 2.2 Different types of pillar extraction methods depending upon the roof condition, spitting and slicing , slicing, extraction , partial extraction, formation of Goaf, withdrawal of support. 2.3 Caving, definition of local fall, main fall, cavability index, dangers associated with caving, precautiod air blast, Hydraulic Stowing, definition of hydraulic profile, (H/L Ratio), Surface and underground arrangements to be done for carrying hydraulic stowing.	Model Demonstration Video Demonstrations Lecture Using Chalk-Board Site/Industry Visit

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Describe the procedure of working of coal seam in Board and Pillar Method. TLO 3.2 Follow safety precautions while working near restricted areas. TLO 3.3 Follow safety precautions while Working near fire area. TLO 3.4 Follow safety precautions while working below waterlogged area. TLO 3.5 Follow safety precautions while working below depillared goaf .	**Unit - III Thick seam and Contiguous Seam Working** 3.1 Thick seam and Contiguous Seam Working, precautions to be taken while working contiguous seams. Layout and case study. 3.2 Precautions while working near restricted areas as per CMR 2017. 3.3 Working near fire area as per CMR 2017. 3.4 Working below waterlogged area as per CMR 2017. 3.5 Working below depillared goaf as per CMR 2017.	Model Demonstration Video Demonstrations Presentations Site/Industry Visit

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Design the longwall Panel under given geological condition. TLO 4.2 Describe different types of longwall mining methods.	**Unit - IV Longwall Mining** 4.1 Applicability of longwall mining, Design of long wall panel, Factors affecting length of longwall face, Barrier width, gate road length. 4.2 Different types of Longwall Mining, Longwall advancing, Longwall retreating ,Cyclic longwall ,Non cyclic long wall, single unit face and double unit face, different cutting machines used Single ended ranging drum shearer (SERD) and Double ended ranging drum shearer (DERD) Layout of DERD manpower calculation.	Video Demonstrations Presentations Lecture Using Chalk-Board Site/Industry Visit
5	TLO 5.1 Describe the theories of subsidence. TLO 5.2 Identify critical factors which causes subsidence. TLO 5.3 Elaborate precautions to be taken for reducing subsidence. TLO 5.4 Explain the techniques of measurement of subsidence.	**Unit - V Subsidence due to Underground Mining** 5.1 Different Theories of subsidence, Normal Theory, Vertical Theory, in-between Normal and vertical, Dome theory, Different definition related to subsidence, different types of failure. 5.2 Different factors affecting subsidence, Depth Thickness, nature of strata, method of working, relation of subsidence with time. 5.3 Precautions to be taken on surface and underground to reduce subsidence Concept of Harmless Depth. 5.4 Measurement of subsidence and preservation of record, various techniques used, Measurement of Subsidence using Survey.	Model Demonstration Video Demonstrations Case Study Lecture Using Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Workout the percentage of extraction during development.	1	Calculation of the percentage of extraction during development under given geological conditions of mining.	2	CO1
LLO 2.1 Prepare layout of Development by Side Discharge Loader (SDL)/Load Haul Dumper (LHD) with chain conveyor.	2	Drawing layout of development by SDL/LHD with chain Conveyor. and Calculate Manpower and Output per Men per Shift (OMS) by assuming your own Conditions.	4	CO1
LLO 3.1 Prepare a layout of Development by Continuous Miner with shuttle cars.	3	Drawing layout of development by Continuous Miner with shuttle cars and calculate Manpower and OMS by assuming your own Conditions.	4	CO1
LLO 4.1 Estimate the number of pillars in closed panel.	4	Calculatation of number of pillars in closed panel under given conditions.	2	CO1
LLO 5.1 Draw Ordinary Isolation stopping and Explosion proof isolation stopping.	5	Creation of Ordinary Isolation stopping and Explosion proof isolation stopping.	2	CO1
LLO 6.1 Evolve suitable arrangements to be made before starting depillaring.	6	Preparation of panel for depillaring. Suggest the suitable arrangements to be done before starting depillaring.	2	CO2
LLO 7.1 Plan the line of extraction.	7	Design of line of extraction under given roof conditions and other factors.	2	CO2
LLO 8.1 Suggest suitable Strata Control and Monitor Plan.	8	Creation of Strata Control and Monitor Plan under given conditions.	2	CO2
LLO 9.1 Prepare the layout of different types of extraction of coal pillars.	9	Drawing of layout of different types of extraction of coal pillars under given conditions.	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 10.1 Prepare layout of Sand Stowing.	10	Drawing of layout of Sand Stowing in underground coal mines.	2	CO2
LLO 11.1 Identify the dangers associated with caving method. LLO 11.2 Suggest suitable remedial measures.	11	Identification of dangers associated with caving method and suggest suitable remedial measures.	2	CO2
LLO 12.1 Draw layout of development of contiguous working. LLO 12.2 Prepare layout of depillaring of contiguous working.	12	Drawing layout of development of contiguous working and Drawing layout of depillaring of contiguous working.	4	CO3
LLO 13.1 Devlope the working plan for working near restricted areas.	13	Preparation of working plan for working near restricted areas.	2	CO3
LLO 14.1 Execute the working plan for Working near fire area.	14	Preparation of working plan for Working near fire area.	2	CO3
LLO 15.1 Ensure the working plan for Working below waterlogged area.	15	Preparation of working plan for Working below waterlogged area.	2	CO3
LLO 16.1 Ensure the working plan for Working below depillared goaf.	16	Devlopement of working plan for Working below depillared goaf.	2	CO3
LLO 17.1 Sketch a layout of Longwall Advancing. LLO 17.2 Sketch a layout of Longwall retreating.	17	Drawing a layout of Longwall Advancing and layout of Longwall retreating.	4	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 18.1 Sketch a layout of Longwall Single unit Face. LLO 18.2 Sketch a layout of Longwall Double unit Face.	18	Draw a layout of Longwall Single unit Face and. Draw a layout of Longwall Double unit Face.	4	CO4
LLO 19.1 Prepare a layout of Non cyclic Longwall Mining using DERD. LLO 19.2 Calculate manpower and OMS for prepared layout.	19	Drawing a layout of Non cyclic Longwall Mining using DERD and Calculatation of manpower and OMS.	4	CO4
LLO 20.1 Sketch Angle of Draw , Limit Line of subsidence basin. LLO 20.2 Sketch Normal plane ,vertical plane of subsidence basin.	20	Interpretion of different theories of Subsidence. and Drawing a Angle of Draw , Limit line , Normal plane ,vertical plane.	4	CO5
LLO 21.1 Sketch a vertical, angle of break. LLO 21.2 Sketch mode of failure , subsidence basin.	21	Interpretion of different terminologies used in Subsidence. and Drawing a vertical, angle of break. and drawing mode of failure of subsidence basin.	2	CO5
LLO 22.1 Identify different types of methods used for measurement of subsidence.	22	Selection of different types of methods used for measurement of subsidence.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Visit any underground coal mine and Draw the layout of Depillaring and calculate OMS.
- Prepare PPT and brief report of histroy of Mining activities in ancient India.
- Visit any underground coal mine and Draw the layout of development and calculate OMS.

Assignment

- Collect the data of nearby mines where DERD/SERD is using for production.
- Collect the information of Subsidence (Case Study).
- Collect the information of nearby mines working with long wall mining methods, machineries, supports, production etc
- Articulate case study of Thick Seam Working in any underground coal mine.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Model of Coal Pillar showing extraction of gallery	1
2	Model showing Hydraulic sand stowing in depillaring area	10
3	Model showing caving method with safety	11
4	Model Showing working of Contiguous seam	12
5	Model showing plan of working below depillared goaf area.	16
6	Model showing Longwall Advancing method.	17
7	Model showing Longwall Retreating method.	17
8	Model showing working of Single unit and double unit Long wall method.	18
9	Smart Classroom fitted with computer and DLP	19,20,21,22
10	Model of layout of Development district by using SDL/LHD	2

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
11	Model of layout of Development district by using Continuous Miner	3
12	Model showing closed panel system of working	4
13	Model of Ordinary and Explosion proof Isolation Stopping	5
14	Model of depillaring district.	6
15	Model showing different Line of Extraction methods.	7
16	Model showing Application of Strata Contrl and Monitoring Plan in depillaring district	8
17	Model showing Stooking, splitting & Slicing of pillar Extraction.	9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Board and Pillar Method (Development)	CO1	20	2	8	6	16
2	II	Board and Pillar Method (Depillaring)	CO2	20	2	8	6	16
3	III	Thick seam and Contiguous Seam Working	CO3	12	2	4	4	10
4	IV	Longwall Mining	CO4	24	2	6	12	20
5	V	Subsidence due to Underground Mining	CO5	14	2	2	4	8
Grand Total				**90**	**10**	**28**	**32**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- First Progressive Test

- Second Progressive test
- Lab Performance
- Journal Submission

Summative Assessment (Assessment of Learning)

- End Year Examination
- Viva -voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	2	2	1			2			
CO2	3	2	3	1			1			
CO3	3	2	1	1			3			
CO4	3	2	1	1			2			
CO5	3	2	1	1			1			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	R. D. Singh	Principles & Practices of Modern Coal Mining	New Age International (P) Limited, Publishers, New Delhi ISBN: 81-224-0974-1
2	Heartman L. Howard	Introductory Mining Engineering	John Wiley& Sons, New York, ISBN: 0-471-62804-2
3	G.K. Pradhan	Explosive and Blasting Technology	Lovely Prakashan, Dhanbad
4	L.C. Kaku	Mining Digest	Lovely Prakashan, Dhanbad
5	CIMFR	Ground Control in Board and Pillar Mines	Lovely Prakashan, Dhanbad
6	Dr B.P. Verma	Roof Bolting	Lovely Prakashan, Dhanbad
7	D.J.Deshmukh	Elements of Mining Technology	Lovely Prakashan, Dhanbad

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.youtube.com/watch?v=oDduxw9d6lw	videos regarding mining
2	https://discoveryjournals.org/discoveryengineering/index.htm	videos regarding mining
3	https://www.slideshare.net/pramodgpramod/coal-mining-methods-74267944?from_search=0	coal minig methods
4	https://youtu.be/wjNOG36i0Aw?si=Y9TXqLiEgApcWHKF	subsidence

Sr.No	Link / Portal	Description
5	https://www.slideshare.net/anilitbhu/longwall-top-coal-caving-method?from_search=7	longwall top coal caving method
6	https://www.slideshare.net/shanz2u/longwall-mining-14602171?from_search=0	longwall mining

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

APPLIED MECHANICS **Course Code : 332301**

Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: APPLIED MECHANICS**
Course Code	**: 332301**

I. RATIONALE

The analysis of forces acting on various structural and machine elements using principles of mechanics gives useful data for detailing and design of structure and machine. The analysis of forces helps to prevent arises of defects, errors and subsequent failures in such elements under action of forces. This course is designed for diploma aspirants to acquire and apply the basic and discipline knowledge to solve practical problems regarding design and automation components in the field of various engineering disciplines such as mining their allied disciplines.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Apply the principles of engineering mechanics to analyze, design and automation the prototypes and equipment's of mining industry

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Select the suitable machine under given in mining condition.
- CO2 - Analyze the given force system to calculate resultant force in mining.
- CO3 - Determine unknown force(s) of given load combinations in the given situation.
- CO4 - Apply the laws of friction in the given situation.
- CO5 - Determine the centroid/center of gravity of the given structural elements of having specific shape and size.

- CO6 - Determine various material parameters under various load conditions.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332301	APPLIED MECHANICS	AMC	AEC	2	-	2	-	4	4	3	30	70	100	40	25	10	25@	10	-	-	150

Total IKS Hrs for Sem. : 1 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Identify the type of machine based on efficiency of machine. TLO 1.2 Calculate effort required and load lifted by the given simple lifting machine. TLO 1.3 Verify law machine for given loading. TLO 1.4 Determine effort required along with efficiency for given machine with varying velocity ratio.	**Unit - I Simple Lifting Machine** 1.1 Concept of simple lifting machine, load, effort, mechanical advantage, velocity ratio, efficiency of machines, reversible and non-reversible/self locking machines. (IKS*: Hand axe as wedge, Lever in battle, Inclined Plane for loading, Pulleys to lift water in irrigation) 1.2 Concept of ideal machine and its conditions, machine friction, ideal effort, ideal load, effort lost in friction and load lost in friction, maximum mechanical advantage and maximum efficiency. 1.3 Nature of graphs: Load vs. effort, load vs. ideal effort, load vs. MA, load vs. efficiency, Law of machine and its uses. 1.4 Velocity ratios of inclined plane, Differential axle and wheel, Worm and worm wheel, Single purchase and double purchase crab winch, Simple screw jack, Weston's differential pulley block, geared pulley block, two sheave pulley block, three sheave pulley block	Lecture Using Chalk-Board Video Demonstrations Presentations Case Study

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Describe the characteristics of force. TLO 2.2 Calculate the moment of given forces in a force system. TLO 2.3 Suggest the suitable law for the analysis of given force system. TLO 2.4 Determine the components of given force. TLO 2.5 Calculate the resultant force of given force system analytically. TLO 2.6 Calculate the resultant force of given force system graphically.	**Unit - II Analysis of Forces** 2.1 Introduction of Mechanics: Engineering Mechanics, Statics, Dynamics, Kinetics, Kinematics, concept of rigid body, Force: definition, unit, graphical representation, Bow's notation, characteristics,Types of force system 2.2 Moment of force: Definition, unit, sign conventions, couple and its properties. 2.3 Law related to forces: Law of transmissibility of force, Law of polygon of forces, Varignon's theorem of moments, Law of moment, Law of parallelogram of forces. (IKS*:Weighing scale in Mohenjodaro, Harappa 2.4 Resolution of coplanar forces: orthogonal and non orthogonal components of a force. 2.5 Composition of coplanar forces using analytical method. Resultant of collinear, concurrent and nonconcurrent force system. 2.6 Composition of coplanar forces using graphical method. Resultant of concurrent force system and parallel force system consisting of maximum four forces only	Lecture Using Chalk-Board Video Demonstrations Demonstration Collaborative learning Presentations Hands-on

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Draw the Free Body Diagram for given loading in given situation. TLO 3.2 Determine the equilibrant of the given concurrent force system. TLO 3.3 Use Lami's theorem to determine the unknown forces causing equilibrium for given practical situation. TLO 3.4 Identify the type of beam in a given structure. TLO 3.5 Determine reactions in the given type of beam analytically.	**Unit - III Equilibrium of Forces** 3.1 Equilibrium and its conditions. 3.2 Equilibrant and relation with resultant, Equilibrant of concurrent force system. 3.3 Lami's Theorem and its applications, Concept of Free body diagram, (Problems unknown not more than two.) 3.4 Types of supports: fixed, simple, hinged, roller and fixed, Types of beams: cantilever, simply supported, overhanging, continuous and fixed. Types of loads: vertical and inclined point load, uniformly distributed load. 3.5 Determination of Beam reactions using analytical method for cantilever simply supported and overhanging beam subjected to vertical load, inclined load and uniformly distributed load (combination of any two types).	Lecture Using Chalk-Board Video Demonstrations Demonstration Presentations Site/Industry Visit Hands-on Case Study

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Determine friction force along with coefficient of friction for the given condition. TLO 4.2 Describe the conditions for friction for the give situation. TLO 4.3 Determine friction force in the given situation. TLO 4.4 Draw free body diagram for showing forces acting on a ladder under given condition.	**Unit - IV Friction of Forces** 4.1 Friction and its relevance in engineering, types and laws of friction, limiting equilibrium, limiting friction, co-efficient of friction, angle of friction, angle of repose, and their relationship. 4.2 Equilibrium of bodies on level surface subjected to force parallel to plane and inclined to plane. 4.3 Equilibrium of bodies on inclined plane subjected to force parallel to the plane only. 4.4 Forces acting on ladder (only free body diagram, no numerical).	Lecture Using Chalk-Board Video Demonstrations Demonstration Presentations Case Study Hands-on Site/Industry Visit
5	TLO 5.1 Determine the centroid of given plane figure. TLO 5.2 Determine the centroid of given composite figure. TLO 5.3 Determine center of gravity of given solid. TLO 5.4 Determine Centre of gravity of the given composite solid.	**Unit - V Centroid and Centre of Gravity** 5.1 Centroid of geometrical plane figures: square, rectangle, triangle, circle, semi-circle, quarter circle (IKS*: Archery arrowheads in Ramayana, Arch in archeological structures such as Mahal, Gol Gumbaz). 5.2 Centroid of composite figures such as I, L, C, T, Z sections (not more than three simple figures). 5.3 Centre of Gravity of simple solids: cube, cuboid, cylinder, cone, sphere and hemisphere (no hollow solids). 5.4 Centre of Gravity of composite solids composed of not more than two simple solids.	Lecture Using Chalk-Board Demonstration Video Demonstrations Model Demonstration Hands-on Case Study

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
6	TLO 6.1 Describe given terms and compute their values. TLO 6.2 Calculate Material Properties Under given load condition. TLO 6.3 Compute stresses and load shared by given composite sections subject to direct load. TLO 6.4 Calculate the change in dimensions and volume of the body subjected to uniaxial, biaxial and tri-axial loads/stresses of given materials. TLO 6.5 Calculate design constants for given conditions. TLO 6.6 Describe relation between given modulus.	**Unit - VI Simple Stress, Strain and Elastic constants** 6.1 Stress, strain, elasticity. Rigid, elastic and plastic bodies. Deformation of elastic body under various forces. 6.2 Type of stresses-normal and tangential, normal tensile and compressive stresses, standard stress strain curve for mild steel bar under tension test, Hook’s law, elastic limit, modulus of elasticity, factor of safety, yield and working stresses. 6.3 Stress and strain in composite section under axial loading. 6.4 Longitudinal and Lateral strain, Poisson’s ratio, biaxial and triaxial stresses, volumetric strain, change in volume, Bulk modulus. 6.5 Shear stress and strain, modulus of rigidity, complementary shear stress and induced tension and compression due to it. 6.6 Relation between modulus of elasticity, modulus of rigidity and bulk modulus.	Lecture Using Chalk-Board Video Demonstrations Cooperative Learning Model Demonstration Hands-on

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs

APPLIED MECHANICS **Course Code : 332301**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Verify law of machine under the given condition. LLO 1.2 Verify law of moment of forces. LLO 1.3 Understand the centroid of structural component.	1	Collect the photographic information of Indian Knowledge System (IKS) given in various units.	6	CO1 CO2 CO5
LLO 2.1 Verify law of machine under the given condition.	2	*Determine mechanical advantage and velocity ratio of differential axle and wheel for different load and efforts.	2	CO1
LLO 3.1 Verify law of machine under the given condition.	3	Determine mechanical advantage and velocity ratio of worm and worm wheel for different load and efforts.	2	CO1
LLO 4.1 Verify law of machine under the given condition.	4	Determine mechanical advantage and velocity ratio of single purchase crab winch for different load and efforts.	2	CO1
LLO 5.1 Verify law of machine under the given condition.	5	Determine mechanical advantage and velocity ratio of double purchase crab winch for different load and efforts.	2	CO1
LLO 6.1 Verify law of machine under the given condition.	6	*Determine mechanical advantage and velocity ratio of simple screw jack for different load and efforts.	2	CO1
LLO 7.1 Verify law of machine under the given condition.	7	Determine mechanical advantage and velocity ratio of Weston’s differential pulley block for different load and efforts.	2	CO1
LLO 8.1 Verify law of machine under the given condition.	8	Determine mechanical advantage and velocity ratio of geared pulley block for different load and efforts.	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Verify law of machine under the given condition.	9	Determine mechanical advantage and velocity ratio of two sheave pulley block for different load and efforts.	2	CO1
LLO 10.1 Verify law of machine under the given condition.	10	Determine mechanical advantage and velocity ratio of three sheave pulley block for different load and efforts.	2	CO1
LLO 11.1 Analyse the resultant force of given force system.	11	*Verify law of polygon of forces using Universal force table for given forces.	2	CO2
LLO 12.1 Analyse the resultant force of given force system.	12	*Verify law of moment of forces using law of moment apparatus for given forces.	2	CO2
LLO 13.1 Analyse the resultant force of given force system.	13	Verify Varignon’s theorem of moments of forces using law of moment apparatus for given forces.	2	CO2
LLO 14.1 Analyse the resultant force of given force system.	14	Determine graphically the resultant force of given concurrent force system.	2	CO2
LLO 15.1 Analyse the given force system acting on structural element.	15	Determine graphically the resultant force of given parallel force system.	2	CO2
LLO 16.1 Verify laws of friction related to forces.	16	*Verify the Lamis theorem using Universal force table apparatus for given forces.	2	CO3
LLO 17.1 Verify laws of friction related to forces.	17	*Determine support reactions of simply supported beam using parallel force or beam reaction apparatus for given vertical forces.	2	CO3
LLO 18.1 Apply the concept of friction for given objects.	18	*Determine coefficient of friction using friction apparatus for given block on horizontal plane.	2	CO4
LLO 19.1 Verify laws of friction related to forces.	19	Determine coefficient of friction using friction apparatus for given block on inclined plane.	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 20.1 Apply the concept of centroid for given objects.	20	Apply the concept of centroid for given objects.	4	CO5
LLO 21.1 To find the sliding coefficient of friction between the belt and pulley LLO 21.2 Apoply Universal Testing Machine for different types of test.	21	Operate Universal Testing Machine for different types of test.	2	CO6
LLO 22.1 understand the elongation of steel by using Extensometer.	22	Measure the elongation of steel by using Extensometer.	2	CO6
LLO 23.1 Apply Compression testing machine for compression test.	23	Operate Compression testing machine for compression test.	2	CO6

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Assignment

- *Solve the examples on calculation of values of MA, VR, ?, Pi, Pf, Wi, Wf etc. for given type of machine. *Solve the examples on calculation of centroid of simple/composite plane figures from given problem statement or figure. *Solve the examples on calculation of centroid of simple/composite plane figures from given problem statement or figure. *Solve the examples on calculation of centroid of simple/composite solid bodies from given problem statement or figure. *Solve the examples on calculation of coefficient of friction, normal reaction, force required to pull the block for given case of frictional bodies (horizontal or inclined plane). *Solve the examples on calculation of unknown forces using Lamis theorem from given problem statement or figure. *Solve the examples on calculation of support reactions of given beam from given problem statement or figure. *Solve the examples on calculation of orthogonal or non orthogonal components of a force. *Solve the examples on calculation of resultant of a force for given force system from given problem statement or figure. *Solve the examples on calculation of moments of a force from given problem statement or figure.

Micro project

- *Prepare a chart showing comparison of centroid and center of gravity for square-cube, rectangle- cylinder, triangle-cone, circle-sphere, semicircle-hemisphere. *Prepare chart of types of forces showing real-life examples. *Prepare chart or flex of laws related to engineering mechanics like law of moment, law of machine, law of parallelogram of forces, Varignon's theorem of moments etc.. *Collect photographs of specific simple lifting machine and relate these machines with the machines being studied and prepare models of simple lifting machines using tools in "MECHANO" and "MECHANIX" *Prepare chart showing all types of beams having types of support (roller, hinged, fixed) with sketches and corresponding photographs of real life examples. *Prepare models of types of beam subjected to all loads (point load, udl, uvl, moment, couple) with sketches and corresponding photographs of real life examples. *Prepare photographic chart showing real life examples of uses of friction on horizontal (walking, writing, etc.) and inclined plane (slider in gardens, loading of heavy material in trucks etc.).

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Simple axle and wheel (wall mounted unit with the wheel of 40 cm diameter and axles are insteps of 20 cm and 10 cm reducing diameter .	1
2	Law of moment’s apparatus consisting of a stainless steel graduated beam 12.5 mm square in section, 1m long, pivoted at centre.	10,11
3	Beam Reaction apparatus (The apparatus is with two circular dial type 10 kg.)	15
4	Friction apparatus for motion along horizontal and inclined plane (base to which a sector with graduated arc and vertical scale is provided. The plane may be clamped at any angle up to 45 degrees. pan. Two weight boxes (each of 5 gm,10 gm, 2-20 gm, 2-50 gm, 2-100 gm weight)	16,17
5	Models of geometrical figures.	18

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
6	Differential axle and wheel (wall mounted unit with the wheel of 40 cm diameter and axles are insteps of 20 cm and 10 cm reducing diameter .	2
7	Pulley and flat belt and V belt	21
8	Worm and worm wheel (wall mounted unit with threaded spindle, load drum, effort wheel; with necessary slotted weights, hanger and thread)	3
9	Single Purchase Crab winch (Table mounted heavy cast iron body. The effort wheel is of C.I. material of 25 cm diameter mounted on a shaft of about 40mm dia. On the same shaft a geared wheel of 15 cm dia.	4
10	Double Purchase Crab winch (Having assembly same as above but with double set of gearing arrangement.)	5
11	Simple screw Jack (Table mounted metallic body , screw with a pitch of 5 mm carrying a double flanged turn table of 20 cm diameter.	6
12	Weston’s Differential pulley block (consisting of two pulleys; one bigger and other smaller.	7
13	Weston’s Differential worm geared pulley block (Consists of a metallic (preferably steel) cogged wheel of about 20 cm along with a protruded load drum of 10 cm dia. to suspend the weights of 10 kg, 20 kg-2 weights and a 50 kg weights)	8
14	Universal Force Table (Consists of a circular 40 cm dia. Aluminum disc, graduated into 360 degrees.) with all accessories.	9,14

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Simple Lifting Machine	CO1	10	2	2	6	10

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
2	II	Analysis of Forces	CO2	14	2	4	8	14
3	III	Equilibrium of Forces	CO3	10	2	8	4	14
4	IV	Friction of Forces	CO4	8	2	4	6	12
5	V	Centroid and Centre of Gravity	CO5	10	2	4	6	12
6	VI	Simple Stress, Strain and Elastic constants	CO6	8	2	2	4	8
Grand Total				**60**	**12**	**24**	**34**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Term work (Lab Manual), Self-Learning (Assignment) Question and Answers in class room, quiz and group discussion.

Note: Each practical will be assessed considering-60% weightage to process related and 40 % weightage to product related

Summative Assessment (Assessment of Learning)

- Practical Examination, Oral Examination, Pen and Paper Test.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	1	1	1	2	1	-	1			
CO2	1	1	1	2	1	-	1			
CO3	2	2	1	2	1	-	1			
CO4	2	2	2	2	1	-	1			
CO5	2	2	1	2	1	-	1			
CO6	1	2	1	2	1	-	1			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	S. Ramamrutham	Engineering Mechanics	Dhanpat Rai Publishing Co. 2016 ISBN-13: 978- 9352164271
2	R. S. Khurmi, N.Khurmi	Engineering Mechanics	S.Chand & Co. New Delhi 2018 ISBN: 978-9352833962

Sr.No	Author	Title	Publisher with ISBN Number
3	S. S. Bhavikatti	Engineering Mechanics	New Age International Private Limited, Edition 5, ISBN: 978-9388818698, New Delhi
4	D. S. Bedi, M. P. Poonia	Engineering Mechanics	Khanna Publishing ISBN-13:978-9386173263, New Delhi
5	Dr. R. K. Bansal	Engineering Mechanics	Laxmi Publications, Darya ganj, Delhi, Edition 6, ISBN 13: 9788131804094

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.engineersrail.com/simple-lifting-machine/	Introduction of simple lifting machine
2	https://www.youtube.com/watch?v=kNypk8GReqM	Law of machine and types of machines useful in industry.
3	https://www.youtube.com/watch?v=6u_rjLjv-MY&list=PLOSWwFV98r	Introduction of force system with examples
4	https://www.youtube.com/watch?v=RrVuSbCZH8c	Verification of Lamis theorem in laboratory
5	https://www.youtube.com/watch?v=tM5hsUiNpGA	Calculation of beam reactions for various types of beams
6	https://www.youtube.com/watch?v=RGT1g_lu440	Calculation of coefficient of friction for horizontal and inclined plane
7	https://www.youtube.com/watch?v=wfjLNSfPXAI	Centroid of plane/composite figures, C.G. of plane/composite solids
8	https://www.youtube.com/watch?v=v6VTMwxx4oA	Centroid of composite figures.

Sr.No	Link / Portal	Description

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: BASIC SURVEYING**
Course Code	**: 332302**

I. RATIONALE

Surveying is generally used to make land maps and boundaries. The development of engineering survey is the basic foundation to ensure the quality of the project, because it can provide accurate data for the subsequent construction. Surveying is involved in everything right from accurately drawing boundaries between private and public land, to inspecting bridges and other critical infrastructure. Without surveying, the placement, security, and safety of projects cannot be assured. Therefore, the students are required to develop such competency to carry out the given type of survey using relevant equipment's so as to prepare the plan to interpret the information to take the appropriate decisions. This course will help the students in achieving in above mentioned goal.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Prepare plans and Contour maps using Surveying Equipment's and Techniques.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Suggest relevant type of survey required for the given situation.
- CO2 - Undertake cross staff and compass survey for the given field.
- CO3 - Undertake survey using Theodolite for preparing a plan of the given terrain.
- CO4 - Determine Reduced Level to prepare Contour maps for the given type of terrain.

- CO5 - Prepare the plan using Plane Table Surveying to locate relevant details.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332302	BASIC SURVEYING	BSU	SEC	2	-	2	-	4	4	3	30	70	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 1 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the given basic principles of surveying. TLO 1.2 Classify the survey based on purpose, instruments used and nature of field. TLO 1.3 Use the conventional sign and symbols for preparing the plan of a given land.	**Unit - I Overview and Classification of Surveying** 1.1 Surveying: Introduction, Purpose, use and Principles. 1.2 Types of surveying- Primary and Secondary classification, Plane, Geodetic, Cadastral, Hydrographic, Photogrammetry Aerial, Layout survey, Control survey, Topographical survey, Route survey, Reconnaissance survey. 1.3 Conventional sign and symbols	Demonstration Lecture Using Chalk-Board Video Demonstrations Presentations Hands-on

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Describe the procedure of finding the distance between two given inter-visible and invisible survey stations. TLO 2.2 Explain the given Survey line and survey station used in survey. TLO 2.3 Explain the methods of ranging. TLO 2.4 Calculate the area of open field using chain and cross staff survey. TLO 2.5 Define Geographic/True Magnetic and Arbitrary Meridians and Bearings, Meridian and Bearing. TLO 2.6 Convert the Whole circle bearing to reduced bearing system and vise versa. TLO 2.7 Calculate internal and external angle from bearing of line. TLO 2.8 Determine the correct bearing from given Data. TLO 2.9 Apply Bowditch's rule to complete the traverse of given land.	**Unit - II Cross Staff and Compass Surveying** 2.1 Linear Measurement Instruments: Metric Chain, Tapes, Arrow, Ranging rod, Open cross staff (IKS). 2.2 Chain survey Station, Base line, Check line, Tie line, Offset, Tie station, Types of offsets: Perpendicular and Oblique. 2.3 Ranging: Direct and Indirect Ranging. 2.4 Area Calculations of field by cross staff (Numerical problems). 2.5 Compass Traversing: open, closed. 2.6 Technical Terms: Geographic/True Magnetic and Arbitrary Meridians and Bearings, Meridian and Bearing. 2.7 Whole Circle Bearing System and Reduced Bearing System . Numericals on conversion of given bearing to another bearing (from one form to another), Fore Bearing and Back Bearing. 2.8 Calculation of internal and external angles from bearings at a station. 2.9 Components of Prismatic Compass and their Functions (No sketch) Temporary adjustments and observing bearings.	Demonstration Presentations Lecture Using Chalk-Board Video Demonstrations Hands-on

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
		2.10 Local attraction, Methods of correction of observed bearings-Correction at station and correction to included angles. 2.11 Methods of plotting a traverse and closing error, Graphical adjustment of closing error.	

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Explain the given components of a transit Theodolite. TLO 3.2 Explain the relationship between the given fundamental axis of theodolite along with typical characteristics. TLO 3.3 Describe the procedure to measure the horizontal Angle using Theodolite for the given situation. TLO 3.4 Describe the procedure to measure vertical angles using Theodolite for the given situation. TLO 3.5 Compute Latitude, Departure, Consecutive co ordinates. Independent coordinates from the data given. TLO 3.6 Determine the type of traverse by undertaking relevant check in the given situation. TLO 3.7 Calculate the bearing from given angles. TLO 3.8 Apply Bowditch's rule along with Transit rule to balance the traverse for a given data. TLO 3.9 Prepare Gale's Traverse table for the given data.	**Unit - III Theodolite Surveying** 3.1 Types and uses of Theodolite; Component parts of transit Theodolite and their functions, Reading the Vernier of transit Theodolite. 3.2 Technical terms- Swinging, Transiting, Face left, Face right. 3.3 Fundamental axes of transit Theodolite and their relationship. 3.4 Temporary adjustment of transit Theodolite. 3.5 Measurement of horizontal angle Direct and Repetition method, Errors eliminated by method of repetition. 3.6 Measurement of vertical Angle. 3.7 Theodolite traversing by included angle method and deflection angle method. 3.8 Checks for open and closed traverse, Calculations of bearing from angles. 3.9 Traverse computation-Latitude, Departure, Consecutive coordinates, independent coordinates, Balancing the traverse by Bowditch's rule and Transit rule, Gale's Traverse table computation.	Demonstration Flipped Classroom Lecture Using Chalk-Board Hands-on Site/Industry Visit Collaborative learning Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain the terms Level surfaces, level line, Horizontal and vertical surfaces, Datum, Bench Marks- GTS, Permanent, Arbitrary and Temporary, Reduced Level, Line of collimation, Back sight, Fore sight, intermediate sight, Change point, Height of instruments. TLO 4.2 Explain the Construction of given levelling equipment with its sailent features. TLO 4.3 Explain the temporary adjustments of dumpy level. TLO 4.4 Calculate Reduced Level of the given station using relevant method of surveying. TLO 4.5 Justify the relevant types of levelling with examples. TLO 4.6 Interpret the contour maps for the given type of topography. TLO 4.7 Describe the characteristics of contours for the given terrain.	**Unit - IV Levelling and Contouring** 4.1 Terminologies: Level surfaces, level line, Horizontal and vertical surfaces, Datum, Bench Marks- GTS, Permanent, Arbitrary and Temporary, Reduced Level, Line of collimation, Back sight, Fore sight, intermediate sight, Change point, Height of instruments. 4.2 ypes of levels: Dumpy, Auto level, Digital level, Fundamental axis of Dumpy Level . Temporary adjustments of Level. 4.3 Types of Levelling Staffs: Self-reading staff and Target staff. 4.4 Reduced level by Plane of collimation method and Rise/ Fall Method. 4.5 Find the R. L. by H.I. method with necessary checks (Numerical problems). 4.6 Find the R.L by Rise and Fall method with necessary checks. (Numerical problems). 4.7 Types of Leveling : Simple, Differential, Fly, Profile and Reciprocal Levelling. 4.8 Contour, contour interval, horizontal equivalent. 4.9 Contour maps: Characteristics and uses of Contour maps.	Demonstration Video Demonstrations Lecture Using Chalk-Board Presentations Cooperative Learning Hands-on

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
		4.10 Methods of Locating Contour: Direct and Indirect.	
5	TLO 5.1 Explain the functions and use of the given type of accessories of a plane table. TLO 5.2 Describe the method of orienting the plane table in a given situation. TLO 5.3 Select the relevant method of plane tabling for a given situation.	**Unit - V Plane Table Surveying** 5.1 Principle of plane table survey. 5.2 Accessories of plane table and their use, Telescopic alidade. 5.3 Setting of plane table; Orientation of plane table - Back sighting and Magnetic meridian method. 5.4 Methods of plane table surveys : Radiation, Intersection and Traversing. 5.5 Merits and demerits of plane table survey.	Demonstration Presentations Lecture Using Chalk-Board Hands-on Video Demonstrations

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Find the distance between two given inter-visible points.	1	*Measure the distance between two intervisible survey stations using chain, tape and ranging rods.	2	CO2
LLO 2.1 Undertake chain and cross staff survey for the given plot.	2	*Determine area of open field using chain and cross staff survey.	2	CO2
LLO 3.1 Calculate area of irregular plot from given plan of plot.	3	Determine area of irregular field using Digital Planimeter .	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 4.1 Determine bearing using Prismatic Compass.	4	*Measure Fore Bearing and Back Bearing of survey lines of open traverse using Prismatic Compass.	2	CO2
LLO 5.1 Prepare traverse using Prismatic Compass.	5	*Measure Fore Bearing and back bearing of a closed traverse of 5 to 6 sides and correct the bearings and included angles.	4	CO2
LLO 6.1 Use transit theodolite to measure Horizontal angle by Direct Method.	6	Measure Horizontal angle by using Transit Theodolite by Direct Method.	2	CO3
LLO 7.1 Use transit theodolite to measure Horizontal angle by method of Repetition.	7	Measure Horizontal angle by using Transit Theodolite by method of Repetition .	4	CO3
LLO 8.1 Use transit theodolite to measure Vertical angle .	8	Measure vertical angle using Transit Theodolite.	4	CO3
LLO 9.1 Prepare traverse using Transit Theodolite.	9	*Use transit theodolite to carry out Survey Project for closed traverse for minimum 5 sides (Compulsory).	6	CO2 CO3
LLO 10.1 Undertake differential leveling by Height of instrument method using dumpy level/Auto Level and leveling staff.	10	*Determine Reduced Level by Height of Instrument Method.	4	CO4
LLO 11.1 Undertake differential leveling by Rise and fall method using dumpy level/Auto Level and leveling staff.	11	*Determine Reduced Level by Rise and Fall Method	4	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 12.1 Undertake fly leveling with double check using dumpy level/ Auto level and leveling staff	12	*Perform Fly Levelling to check levelling work.	2	CO4
LLO 13.1 Perform Road profile and cross section of given terrain.	13	*Profile leveling and cross-sectioning for a road length of 300 m with cross-section at 20 m interval. (Compulsory).	6	CO4
LLO 14.1 Undertake differential levelling operation for agriculture land .	14	Undertake differential leveling by using dumpy level/Auto Level and leveling staff for Installation of irrigation pipelines.	4	CO4
LLO 15.1 Conduct block contouring for the area of 40m x 40m to draw its contour plan.	15	Prepare Contour Plan/map using Block Contouring for the area of 40m x 40m to draw its contour plan.	4	CO4
LLO 16.1 Prepare Contour Plan/map using block contouring method.	16	*Plotting contour map using block contouring method for a block of 150m x 150m with grid of 10m x 10m for given land parcel. (Compulsory).	6	CO4
LLO 17.1 plotting contour map using block contouring method for 10 Acre Agriculture land.	17	Prepare Contour plan for control farming using block contouring method .	2	CO4
LLO 18.1 Use plane table survey to prepare plan and locate details by using Radiation Method.	18	Prepare plans and locate details by using Radiation Method.	2	CO5
LLO 19.1 Use plane table survey to prepare plans and locate details by Intersection Method.	19	*Prepare plans and locate details by Intersection Method.	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 20.1 Use plane table survey to prepare plans locate details by Traversing Method.	20	*Prepare traverse using Plane table Surveying .	4	CO5
LLO 21.1 Use plane table survey to prepare plans plan to establish plant nursery.	21	Prepare plan to establish plant nursery .	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Assignment

- Determine the reservoir capacity from a give contour map of reservoir.
- Measure area of small open ground by plane tabling.
- Measure the height of the flag post using Theodolite.
- Draw the representative contour maps for the following: Ridge of a mountain, Hillock, Valley, Pond/lake, Gentle slope, Very Steep Slope, Plain Surface
- Explain one method each to measure the distance between points on either side of obstacles in case of following: River, Lake, Building.
- Set the alignment of proposed road using Theodolite.

- Interpret the given contour maps.
- Determine the reservoir capacity from a give contour map of reservoir.

Micro project

- Observe Topographical maps and interpret the details.
- Collect the contour maps of different terrains available with various authorities & prepare a report on its interpretation.
- Determine the RLs of the components of existing structures like Plinth, lintels, chajja, slab, and beam etc
- Collect the information of survey instruments available in the market with their specifications.
- Prepare a flex chart to explain one method of plane tabling.
- Compare Traversing with plane table and compass method.
- Perform reconnaissance survey for plotting the alignment of road.
- Carry out comparative study of following survey instruments of different make and brands : Auto level and Dumpy Level
- Collect the map of city /town and calculate the ward wise and total area using digital planimeter.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Arrows 400 mm long and made up of good quality hardened and tempered steel wire of 4 mm in diameter.	1,2
2	Metric Chain made from galvanized mild steel wires 4mm in dia, brass handles with swivel joints, brass tallies provided at every 5 m length of chain - 20 and 30m.	1,2
3	Metallic Ranging rods of 2 m length, circular or octagonal in cross section of 30 mm diameter, Lower shoe of 150 mm long. Painted in black, white and red stripes of 200 mm each.	1,2,3,4,5,6,7,8,9,13,14,15,16,17,18,19,20,21
4	Pegs of length 400 mm and c/s area of 50 mm x 50 mm	1,2,3,4,5,6,7,8,9,18,19,20,21
5	Metallic tape-, Steel tape, Invar, Fiber glass tape satisfying IS 1269 (Part 1 and Part 2) : 1997 specifications.	1,2,5,9,13,14,15,16,17,18,19,20,21
6	Dumpy level and automatic levels confirming to IS: 9613 –1986 with stand and internal focusing telescope of standardmake.	10,11,12,13,14,15,16,17
7	Leveling staff- 2 m and 4 m, telescopic type confirming to IS 11961 -1986 or Folding type confirming to IS 1779 (1961), 5 mm least count.	10,11,12,13,14,15,16,17
8	Plane table with accessories- Plane and telescopic Alidade, Trough compass, U-fork ,Spirit level.	18,19,20,21
9	Digital planimeter of standard make with Ni Cd batteries and AC Adapters	3
10	Prismatic compass confirming to IS 1957-1961 with stand, made in Gun metal material having diameter of 85-110 mm and the least count of 30 minutes.	4,5

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
11	Twenty Second Transit theodolite with accessories.	6,7,8,9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Overview and Classification of Surveying	CO1	6	2	4	0	6
2	II	Cross Staff and Compass Surveying	CO2	12	4	4	6	14
3	III	Theodolite Surveying	CO3	18	4	4	12	20
4	IV	Levelling and Contouring	CO4	18	2	8	12	22
5	V	Plane Table Surveying	CO5	6	4	4	0	8
Grand Total				**60**	**16**	**24**	**30**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Termwork, Assignment, Microproject (60% Weightage to process and 40% weitage to product), Question and Answer

Summative Assessment (Assessment of Learning)

- Pen and Paper Test (Written Test), Practical Exam, Oral Exam

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3						2			
CO2	3	3	1	2	1	1	3			
CO3	3	3	2	3	1	2	3			
CO4	3	3	2	3	1	2	3			
CO5	3	2	2	3	1	2	3			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Kanetkar T. P.; Kulkarni, S. V.	Surveying and Levelling volume I	Pune Vidyarthi Gruh Prakashan, Pune; ISBN:978-81-858-2511-3
2	Basak, N. N.	Surveying and Levelling	McGraw Hill Education, New Delhi ISBN 93-3290-153-8

Sr.No	Author	Title	Publisher with ISBN Number
3	S. K. Duggal	Survey I	McGraw Hill Education, New Delhi, ISBN: 978-00-701-5137-6
4	Punmia, B.C, Jain, Ashok Kumar Jain, Arun Kumar	Surveying I	Laxmi Publications., New Delhi. ISBN: 8- 17-008853-4
5	Bhavikatti, S. S.	Surveying and Levelling, Volume 1	I. K. International, New Delhi ISBN: 978- 81-906-9420-9

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://dspmuranchi.ac.in/pdf/Blog/Survey.pdf	Surveying and Levelling
2	https://archive.nptel.ac.in/courses/105/104/105104101/	Introduction to Surveying, Principles of surveying, and Classification of Surveying
3	https://lnct.ac.in/wp-content/uploads/2020/03/UNIT4B.pdf	Theodolite Surveying.
4	https://www.slideshare.net/gauravhtandon1/plane-tablesurvey -27614680	Plane Table Surveying-accessories and methods
5	http://www.pkace.org/Lecture_Notes/Survey-lecturenotes.pdf	Levelling-methods of levelling and types of levels
6	https://civilplanets.com/compass-surveying/	Compass Surveying and its types, Temporary adjustmments
7	http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1282 85	Traversing by Prismatic Compass, WCB and RB conversion and Terms in Compass Surveying
8	https://www.youtube.com/watch?v=x9ZPMxrlS3U	Measurement of bearing by prismatic compass

Sr.No	Link / Portal	Description
9	https://youtu.be/j8poe2vvD2Q	Temporary adjustment of auto level
10	https://www.youtube.com/watch?v=c9U0xlmCzGI	Temporary adjustment of Transit Theodolite
11	https://youtu.be/L54T4uvpMTg	Levelling operation by using Dumpy Level
12	https://www.youtube.com/watch?v=boPrQFZEn9A	Radiation method by plane table surveying
13	https://www.youtube.com/watch?v=PQfr1LABZWg	Contouring and its characteristics, Methods of Contouring
14	https://www.youtube.com/watch?v=-mkf7uJG8DI	Intersection method of Plane Table Surveying
15	https://theconstructor.org/surveying/chain-survey/29812/	Chain, Tapes and other linear measurement equipments

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

ECONOMIC AND FIELD GEOLOGY **Course Code : 332303**

Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: ECONOMIC AND FIELD GEOLOGY**
Course Code	**: 332303**

I. RATIONALE

This course will help the students to use knowledge of geology, stratigraphy and remote sensing in mining engineering works, geological mapping, in search of ground water and nature of strata. After acquiring knowledge and related skill student will choose site according to its use for mining and civil engineering projects. This course is also useful for the ground control underground mines and slope stability in opencast mining

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help students to achieve following industry identified outcomes through various learning experiences in a) Use geological knowledge and skills to identify Indian geological formation b) Execute knowledge of an exploration and economic mineral, mining work by using conventional and remote sensing method.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Identify Ore minerals, Industrial minerals and Economical minerals.
- CO2 - Use knowledge of Geological Formation of India in given situation.
- CO3 - Apply prospecting and subsurface exploration method in given situation.
- CO4 - llustrate Hydrological characteristics and interpretation of water table contour map.
- CO5 - Analyse Engineering Geological problems and study of remote sensing.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL Practical				Based on SL		Total Marks
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332303	ECONOMIC AND FIELD GEOLOGY	EFG	DSC	2	-	2	1	5	5	3	30	70	100	40	25	10	25#	10	50	20	200

Total IKS Hrs for Sem. : 1 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe given processes of Economic Minerals formation. TLO 1.2 Explain given properties of specified Economic Minerals. TLO 1.3 Describe given physical ,chemical properties and origin of Petroleum. TLO 1.4 Intoduction to Minerals and their Exploitation in Ancient India.	**Unit - I Economic Geology and Petroleum** 1.1 Elements of Economic geology, Process of ore formation of economic Minerals deposits with examples. 1.2 Metaliferrous deposits of India – Iron,Copper,Mangnese,Aluminium,Gold,Lead and Zinc (Fe, Cu, Mg, Al, Au, Pb& Zn)Mineralogy, Origin, Distribution, uses, grades and reserves. 1.3 Petroleum-Physical and Chemical properties,Uses Origin and Occurrence of Petroleum in India. 1.4 Iron age in India.(1200BC- 600 AD) (IKS)	Video Demonstrations Presentations Lecture Using Chalk-Board

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Interpret principles of Stratigraphy and PhysiographIc Divisions of India. TLO 2.2 Describe given Geological Formations of India. TLO 2.3 Introduction to Ancient Geology.	**Unit - II Stratigraphy** 2.1 Principles of Stratigraphy, Geological Time-scale, Physiographic Divisions of India. 2.2 Major geological formations of India.Classification, Distribution, Lithology, Structures and Economic importance of following systems. 1) Archean & Dharwar system 2)Cuddapah system 3)Vindhyan System 4)Gondwana super group 5)Deccan trap 2.3 Minerals and Metals in Harappan era.(3200-1800BC) (IKS)	Demonstration Presentations Case Study Lecture Using Chalk-Board
3	TLO 3.1 Select relevant field equipments for Geological mapping. TLO 3.2 Describe various Geological Prospecting method. TLO 3.3 Use suitable Geological Prospecting method.	**Unit - III Geological Field Work** 3.1 Field equipment's required for geological Mapping, Method of field work, and collection of samples. 3.2 Geological prospecting-Airborne prospecting Method, aerial photographs,Ground prospecting Method,Surface prospecting Method, Mapping,Test pitting, trenching, editing, drilling. 3.3 Geophysical Methods of Prospecting -Magnetic,gravity,electrical, resistivity. 3.4 Geochemical Methods of prospecting-Soil, stream, water, rock sampling methods.	Case Study Demonstration Presentations Lecture Using Chalk-Board

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Describe Ground Water Hydrology. TLO 4.2 Explain various water bearing properties of rocks. TLO 4.3 Interpret water table Contour Map. TLO 4.4 Explain Groundwater Movement. TLO 4.5 Interpret Ground Water Provinces of India.	**Unit - IV Ground Water Hydrology** 4.1 Hydrological cycle, Zones of Ground water, Occurrence of ground water, Geological work done by groundwater. 4.2 Water bearing properties of rocks, Porosity, permeability. 4.3 Water table contour map. 4.4 Ground water monitoring system, Darcy's law of ground water movement, specific yield and specific retention, pumping of well. 4.5 Ground water provinces of India, Interpretation of map, location and types of rocks found in various provinces.	Video Demonstrations Presentations Lecture Using Chalk-Board

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Explain relevant engineering properties of rocks TLO 5.2 Investigate geological problems related to Dams and Tunnels in given situation. TLO 5.3 Describe Remote Sensing and GIS	**Unit - V Engineering Geology and Remote sensing** 5.1 Engineering properties of rocks ,Criteria for selection of building stones- availability, workability, strength, durability, porosity, permeability, texture, resistance the, colour and cost. 5.2 Geological problems associated with dam and tunnel side. 5.3 Introduction to remote sensing techniques ,analog and digital data product. 5.4 Remote sensing satelite and application of remote sensing in mining operation. 5.5 Introduction to GIS and its application.	Video Demonstrations Presentations Case Study Lecture Using Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify any two Iron ore minerals based on given physical properties.	1	*Identification of iron ore minerals (Hematite / Limonite / Magnetite/ Siderite/ Pyrite/ Pyrhotite/ Goethite/ Siderite/ Marcasite.)	2	CO1
LLO 2.1 Identify any two Manganese ore minerals based on given physical properties.	2	*Identification Manganese ore minerals (Braunite / Pyrolusite/ Psilomelane/ Manganite.)	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Determine any two Iron ore minerals based on given physical properties.	3	*Identification of Copper ore minerals (Azurite/ Malachite/ Cuprite/ chalcopyrite/ covellite/ Bornite/ Chalcocite.)	2	CO1
LLO 4.1 identify of anyone mineral based on given physical properties.	4	*Identification Lead and Zinc ore minerals – (Galena, sphalerite)	2	CO1
LLO 5.1 identify any two Aluminium ore minerals based on given physical properties.	5	*Identification of Aluminium ore minerals (Bauxite/corundum/ gibbsite/Correndum/Diosper)	2	CO1
LLO 6.1 Interpret physical properties of any two Miscellaneous ore minerals.	6	*Identification of Miscellaneous ore minerals.(Chromite/ Ilmenite/ Baryte/ Magnesite/ Arsenopyrite/ Realgar/ Monazite/ Molybdnite/ Cinnabar/ Orpiment/ Wolframite.)	2	CO1
LLO 7.1 Select three suitable Industrial minerals based on given physical properties.	7	*Identifification of Various types Industrial minerals (Talc/Kyanite,Graphite/Asbestos/Gypsum/Graphite/Coal/ Limestone/Calcite/Mica/Silliminite/Garnet/Asbestos.)	2	CO1
LLO 8.1 Draw Geological time scale with stratigraphical units.	8	*Indian Statrigraphy	2	CO2
LLO 9.1 Draw Geological succession of Archean and Gondwana rocks.	9	*Geological Formation of India.	2	CO2
LLO 10.1 Interpretation of topography from given topographical maps.	10	*Study of Toposheet (Any 5)	2	CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 11.1 Interpretation of Geology from given Satellite Imagery.	11	*Study of Satellite photographs. (Any three)	2	CO3
LLO 12.1 Draw outcrop map by three point problem method.	12	*Outcrop Map (Three point problem method)	2	CO3
LLO 13.1 Draw outcrop map when true dip is given on the map.	13	*Outcrop Map (True dip is given)	2	CO3
LLO 14.1 Draw outcrop map when out crop exposures are given on the map.	14	*Outcrop Map (Outcrop exposures are given)	2	CO3
LLO 15.1 Draw outcrop map when out crop exposures are given only on one contour.	15	*Outcrop Map (One Counter given)	2	CO3
LLO 16.1 Draw a outcrop map When out crop exposures are given in a bore hole on either side of fault.	16	*Outcrop Map(When out crop exposures are given in a bore hole on either side of fault.)	2	CO3
LLO 17.1 Calculation of strike,dip and depth of given strata,When out crop exposures are given in a bore hole on either side of fault.	17	*Calculation of strike,dip and depth of given strata.	2	CO3
LLO 18.1 Calculate dip and strike of beds by actual measurement in the field by Compass.	18	*Aplication of Brunton/Clinometer Compass.	2	CO3

ECONOMIC AND FIELD GEOLOGY **Course Code : 332303**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 19.1 Calculate apparent dip when true dip is given in amount by graphical method.	19	Calculation of apperent dip when true dip is given .	2	CO3
LLO 20.1 Calculate true dip when apparent dip is given in amount by graphical method.	20	Calculation of true dip when apparent dip is given.	2	CO3
LLO 21.1 Draw a water table map on the basis of given water table data of the area.	21	Water Table Map (By given water table data)	2	CO4
LLO 22.1 Interpretation of Remote Sensing Techniques in ground water survey.	22	Remote Sensing Techniques in ground water survey.	2	CO4
LLO 23.1 Identify various Engineering properties of rocks such as,Porosity, Permeability,Strength,Hardness.	23	Engineering Properties of Rocks .	2	CO4
LLO 24.1 Prepare a report on any one dam sites of Indian dam project.	24	Dam Sites Visit on Indian dam project.	2	CO5
LLO 25.1 Acquire knowledge about ground water monitoring.	25	Ground Water Monitoring	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- 1)Collect photographs and samples of ore minerals by geological field work and prepare report. 2)Interpret the geological succession of nearby area prepare a report. 3)Visit to nearby site of G.S.I. and submit report of prospecting exploration work of the site you visited. 4)Visit to remote sensing station. Collect satellite imaginaries of the earth. 5)Visit to hydrological survey of India or G.S.D.A. submit report of ground water exploration work,photographs. Prepare rain water harvesting model or “jalyuktshiwar yojna”. 6)Visit to nearby dam or road, tunnel or bridge site, check how the stability of structure is checked, collect the samples, photographs and prepare report.

Assignment

- Prepare chart on Geological Time Scale and Geological succession. Indian mining industry in respect of any one metallic deposit. (Iron/Copper/Manganese/Cement/Coal) Prepare chart on Ground Water zone. Prepare report on Geological Field work by visiting near by Geological Camp. Prepare a assignment on Remote Sensing Satellite.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Set of Iron ore minerals.(magnifying lenses, pen knife, streak plate)	1
2	Note-From Sr.No.1-7 must contain at least 2 or 3 sets of minerals,ores and each set contain at least 6 different samples.	1,2,3,4,5,6,7
3	Toposheet of survey of India 4x4 feet (5 Toposheet)	10
4	Colour Satellite images between 2 and 5 meter high resolution.(5 Images)	11,22
5	Set of outcrop maps with mentioned details.(set squares, roller scale, and protractor.)	12,13,14,15,16,17
6	Brunton compass and Clinometer compass (damping time of needle: <14 seconds dial scale : 1° size : 80 x 70 x 35 (mm) weight : 250 grams.), set squares, roller scale, and protractor.	18,19,20
7	Set of Manganese ore minerals.(magnifying lenses, pen knife, streak plate)	2

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
8	Bore hole data of water level.	21
9	UTM,Shear Stress,Compression test.	23
10	Set of Copper ore minerals.(magnifying lenses, pen knife, strike plate)	3
11	Set of Lead and Zinc ore minerals (magnifying lenses, pen knife, streak plate)	4
12	Set of Aluminium ore minerals.(magnifying lenses, pen knife,streak plate)	5
13	Set of Miscellaneous ore minerals. (magnifying lenses, pen knife and streak plates)	6
14	Set Industrial Minerals.(magnifying lenses, pen knife, streak plate)	7
15	Relevant chart and books.	8,9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Economic Geology and Petroleum	CO1	12	2	4	8	14
2	II	Stratigraphy	CO2	14	6	4	6	16
3	III	Geological Field Work	CO3	12	2	6	6	14
4	IV	Ground Water Hydrology	CO4	10	4	4	6	14
5	V	Engineering Geology and Remote sensing	CO5	12	2	4	6	12
Grand Total				**60**	**16**	**22**	**32**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Mid term Test,Rubrics for CO's,Assignment,Self Learning and Term work,Seminar /presentation.

Summative Assessment (Assessment of Learning)

- Lab performance,End of term examination,viva voce
- Lab performance,End of term examination,viva voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3			2	2	1	3			
CO2	3			1	2	1	3			
CO3	3	2		2	3	3	3			
CO4	3	3		2	3	3	3			
CO5	3	3		2	3	3	3			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Parbeen Singh	Engineering and General Geology	S.K. Kataria and Sons, 2013, ISBN-13: 978-9350142677
2	Bangar K.M	Principles of Engineering Geology	Standard Publishers Distributors Delhi, 2013, ISBN-13: 978-8180141157
3	DR.S.KUMAR	BASICS OF REMOTE SENSING AND GIS	UNIVERSITY SCIENCE PRESS ISBN NO.978-81-318-0544-2
4	R.N.P.AROGYASWAMI	COURSES IN MINING GEOLOGY	CBS PUBLISHER ISBN 13-978-8120409378
5	Dr.R.K.Bopche Dr.D.K.Agarwal	General and Engineering Geology	Global Education Limited 2018 ISBN-978-81-935799-2-3
6	A.PARTHSARATHI,V. PANCHAPAKESAN,R. NAGARJUN	ENGINEERING GEOLOGY	WILEY V. B. PUBLICATION NEW DELHI 2003 ISBN 978-8126509461
7	G. B. Mahapatra	A Textbook of Physical Geology	CBS Publishers & Distributors, ISBN 10-8123901100

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.gsi.gov.in/webcenter/portal/OCBIS	It has reports, maps, manuals and Guidelines. Training programmes for learning
2	https://pubs.geoscienceworld.org/	a free, public map-based toolset that allows users to search for cross sections, charts, tables, fig

Sr.No	Link / Portal	Description
3	www.geologyshop.com	printed products including maps, sheet explanations and regional guides.
4	www.geologyshop.com	printed products including maps, sheet explanations and regional guides.
5	www.geology.com	This web site has information and News about Geology and Earth Science
6	https://www.surveyofindia.gov.in/	It is a official website of Survey of India having information about maps, mines and geological info
7	https://www.bgs.ac.uk/discovering-geology/	This is good for enhancing knowledge by watching few relevant programmes on Mines and Mine Survey
8	https://youtube.com/@ExploringGeologybyGeolovers	This video is good on Exploring Geology in Hindi and information about Jobs in India.
9	https://www.mecl.co.in/Index.aspx	Site related to exploration related work.

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: MINING TECHNOLOGY**
Course Code	**: 332305**

I. RATIONALE

The students of Mining and Mine surveying must know basic mining operations carried out in underground Coal and Metal Mines. The knowledge of Drilling, Charging, Explosives and detonators used for underground Blasting is essential for Mining Diploma Holders. As a Overmen /Sirdar diploma holder is in charge of the under ground mining district and responsible for safety of persons working under him . The knowledge of different types of roof supports will help mining engineer to choose appropriate method of supporting to make working safe. The course will also provide useful information to students for shot firing and ground control in underground coal and metal mines.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences: Select relevant type of support for different ground conditions and undertake the blasting operation at face as per regulation.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Choose relevant explosives and accessories used for underground blasting.
- CO2 - Prepare the face for solid blasting in underground mines.
- CO3 - Carry out the process of shot firing safely in underground mines.

- CO4 - Install and withdraw the different types of conventional supports used in underground mines.
- CO5 - Install different types of roof bolt and cable bolts in underground mines.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332305	MINING TECHNOLOGY	MTE	DSC	2	-	2	-	4	4	1.5	30	70*#	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 1 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Classify given types of explosives based on their properties TLO 1.2 Select suitable type of Permitted explosives in given situation TLO 1.3 Select suitable type of detonators in a given situation. TLO 1.4 Choose suitable type of safety fuse, detonating cords, detonating relays in given situation TLO 1.5 Suggest proper types of shot firing tools and exploders for particular application.	**Unit - I Explosives And Blasting Accessories** 1.1 Ancient explosive and blasting process (IKS), Common explosive bases, Properties of Explosives, High Explosive & Low explosive, and their comparison. 1.2 Permitted explosives their types, composition, properties, uses, advantages & disadvantages. Brand names of some commonly used explosive of each type. 1.3 A detonator, common types of detonators, plain detonators, instantaneous and delay action detonators their construction, uses, comparison etc. low tension & high-tension detonators. 1.4 Safety fuses, detonating cords, detonating relays. 1.5 Applications of various Shot firing tools, and exploders	Presentations Lecture Using Chalk-Board Lecture Using Chalk-Board Video Demonstrations Model Demonstration

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Interpret given drilling patterns for shot firing. TLO 2.2 Explain proper sequence of shot firing procedure. TLO 2.3 Follow the precautions while blasting-off-solid. TLO 2.4 Describe Historical Blasting Practices.	**Unit - II Blasting Practices in U/G Mines** 2.1 Drilling patterns for shot firing on machine cut face, in stone drift etc. 2.2 Face preparation for shot firing, Preparation of priming charge, charging of hole in coal and rock in underground working only, Direct and inverse initiation, shot firing circuits, procedure of shot firing of holes in gassy mine, precautions. Simultaneous & delay firing. 2.3 Solid blasting, conditions to be satisfied before doing solid blasting, advantages of solid blasting, drilling patterns used with solid blasting 2.4 The use of explosives in mining in ancient time and the standard method for blasting rocks was to drill a hole to a considerable depth and deposit a charge of gunpowder.	Video Demonstrations Video Demonstrations Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Calculate detonator factor and powder factor. TLO 3.2 Identify causes of Misfire shot. TLO 3.3 Identify stemming materials and their purpose. TLO 3.4 Interpret the explosive magazines. TLO 3.5 Describe the procedure of disposal of out dated explosives.	**Unit - III Safety in Blasting** 3.1 Charge of explosive required for blasting in coal, rock. Powder factor, detonator factor. Precaution to improve blasting results. 3.2 Misfires, causes, remedy and method of relieving dealing with misfires, blown out shots, blown through shots causes and precautions. 3.3 Purpose of stemming, Stemming materials used for shot firing, water ampoules for stemming. 3.4 Storage of explosives, Magazines 3.5 Disposal of outdated explosives.	Lecture Using Chalk-Board Video Demonstrations Presentations Video Demonstrations Video Demonstrations
4	TLO 4.1 Explain various types of support used in Mines. TLO 4.2 Identify given types of wooden support Prop, chock, crossbar, fore poling, safari support. TLO 4.3 Describe the installation and Withdrawal of support. TLO 4.4 Interpret support density as per given support plan.	**Unit - IV Conventional Roof Support** 4.1 Classification of different types of supports used in Underground Mines. 4.2 Different types of Wooden Supports used in Underground Mines (Description of Prop , chock, crossbar, fore poling , safari support, only) 4.3 Installation and Withdrawal of support. 4.4 Support Plan and support density.	Lecture Using Chalk-Board Video Demonstrations Video Demonstrations Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Apply the theory of mechanics of strata behavior for stability of underground opening. TLO 5.2 Describe the use of various types of roof bolt supports based on their principle of action TLO 5.3 Follow the specific code of practice for roof bolting. TLO 5.4 Describe the procedure of roof stitching. TLO 5.5 Describe the procedure of cable Bolting TLO 5.6 Classify given types of yielding props. TLO 5.7 Identifygiven types of friction props and hydraulic props.	**Unit - V Roof Bolting, Cable Bolting and Yielding Prop** 5.1 Theories of mechanics of strata behavior: Dome or arch theory, Beam theory. 5.2 Roof Bolting: Theories, Principle of action, functions, Types (Cement capsule grouted bolts, Resin bolt). 5.3 Anchorage testing of Roof Bolts, Code of practice for roof bolting in underground mines. 5.4 Roof stitching, Principle of Roof stitching. 5.5 Cable Bolting, W-strap. 5.6 Yielding supports, Classification .important definition like setting load, 5.7 Different types of friction prop and Hydraulic Props and their installation and withdrawal.	Presentations Video Demonstrations Presentations Video Demonstrations Video Demonstrations Lecture Using Chalk-Board Model Demonstration

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify different Types of permitted Explosives Cartridges.	1	*Identifcation of permitted Explosive Cartridges. (P1-Permitted Explosive,P2-Sheathed Explosive,P3-Equivalent Sheathed Explosive,P4-Extra Safe Explosive,P5-Explosive of Solid Blasting.	2	CO1
LLO 2.1 Use the cut section of Instantaneous Electric Detonator	2	*Demonstration of cut section of Instantaneous Electric Detonator. (Neoprene Plug,Flashing Mixture,Prime Charge (ASA),Base Charge (PETN).	2	CO1
LLO 3.1 Use the cut section of delay Detonator.	3	*Demonstration of cut section of delay Detonator. . (Neoprene Plug,Flashing Mixture,Delay Element,Prime Charge (ASA),Base Charge (PETN).	2	CO1
LLO 4.1 Use the different types of shot Firing tools.	4	*Identification of different shot Firing tools.(Electric Lamp/Torch, Stop Watch, Flame Safety Lamp, Wooden Stemming Rod, Scraper, Crack Detector, Wooden Pricker , Sharp Knife and Crimper).	2	CO2
LLO 5.1 Use different drilling patterns for shot firing.	5	*Demonstration of different drilling patterns for shot firing. (Concentric Rings, Pyramid Cut, Diamond Cut, Wedge Cut, Fan Cut, Drag Cut, Burn Cut and Coromant Cut.)	2	CO2
LLO 6.1 To Prepare and Understand Prime Cartridge.	6	*Prepare primer cartridge for solid blasting. (Using Explosive Cartridge and Detonator).	2	CO2
LLO 7.1 Identify the direct initiation and indirect initiation of blast hole.	7	*Demonstration of direct initiation and indirect initiation of blast hole (Direct Initiation for coal mines and Indirect Initiation for drivages of shaft, drifts and tunnels).	2	CO2
LLO 8.1 Identify series and parallel connection for blast holes.	8	*Preparation of Blasting Circuits (Series and Parallel).	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Calculate detonator factor and powder factor under given conditions/ problems.	9	*To determine detonator factor and powder factor under given conditions/ problems.	2	CO3
LLO 10.1 Identify Misfire shot and carry out the procedure to deal Misfire shot.	10	*Identification of Misfire shot and carry out the procedure to deal Misfire shot.	2	CO3
LLO 11.1 Draw a layout of Magazine under given conditions.	11	*Preperation of layout of Magazine under given conditions.	2	CO3
LLO 12.1 Install the prop support (with the help of model) under given condition.	12	*Installation of Prop Support.	2	CO4
LLO 13.1 Install the chock/cog support (with the help of model) under given condition.	13	*Installation of Chock/Cog Support.	2	CO4
LLO 14.1 Install the crossbar, and safari support under given condition.	14	*Installation of Cross bar and Safari Support.	2	CO4
LLO 15.1 Carryout the appropriate steps for fore poling method of Support.	15	*Installation of Fore Poling Method of Support	2	CO4
LLO 16.1 Withdraw the support using Sylvester prop withdrawer with the help of model.	16	*Demonstrate the procedure of withdrawal of support.	2	CO4
LLO 17.1 Draw support plan and frame SSR under given conditions.	17	*Preparation of support plan and apply of SSR	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 18.1 Identify different types of roof bolts	18	*Demonstration of Roof Bolts	2	CO4 CO5
LLO 19.1 Develop the procedure to install cement capsule (quick setting cement) roof bolts using video clip and virtual media.	19	Installation of Cement Capsule Roof Bolt	2	CO5
LLO 20.1 Develop the procedure to install resin capsule roof bolts using video clip and virtual media.	20	Installation of Resin Capsule Roof Bolts	2	CO5
LLO 21.1 Develop the procedure to install Roof stitching using video clip and virtual media.	21	Installation of Roof Stitching	2	CO5
LLO 22.1 Develop the procedure to install Cable bolting using video clip and virtual media.	22	Installation of Cable Bolting	2	CO5
LLO 23.1 Install Friction Prop (with the help of model) under given condition.	23	Installation of Friction Prop	2	CO5
LLO 24.1 Install hydraulic prop (with the help of model) under given condition.	24	Installation of Hydraulic Prop	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- •Detonators: Prepare report of various Detonators used in that mine by visiting any underground coal mine.
- Prepare a report on Safety Fuse, Detonating cords and Detonating Relays used in that mine by visiting any underground coal mine.
- Explosives and Blasting Accessories: Prepare report of various explosives and accessories used in that mine by visiting any underground coal mine.
- Prepare a Report on Roof Bolting and Cable Bolting.
- SSR : Collect the copy of Systematic Support Rules of nearby mines and calculate support density.
- Conventional Roof Support : Prepare the cut out Model of Different types of support used in mines.
- Safety in Blasting: Collect actual date for measuring pull, Detonator factor and powder Factor.
- Prepare a report on Prime cartridge ,Direct Initiation and Inverse Initiation .
- Blasting Patterns: Prepare a Report on Various types of Blasting /Drilling Patterns.
- Prepare a Report on various shot firing tools and Exploders used in that mine by visiting any underground coal mine.
- Yielding Support : Collect data on different types of Yielding Support used in Indian Mines ,their Load bearing capacity and manufacturers.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Dummy cartridges of different Types of permitted Explosives Cartridges.	1,6,8
2	Flame safety lamp with dark room facility.	10
3	Models of the prop, chock, crossbar, and safari support.	12,13,14
4	Model of fore poling method of Support	15
5	Sylvester prop withdrawer.	16
6	Model of cement capsule (quick setting cement) used in roof bolts.	18,19
7	Cut out models of Instantaneous Electric Detonator	2
8	Model of different types resin capsule used in roof bolts.	20
9	Model of Friction Prop	23
10	Model of Hydraulic Prop	24

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
11	Cut out models of delay Detonator used for Shot firing in underground Mines.	3
12	Different shot Firing tools.	4
13	Crack detector used for detecting cracks in shot holes.	4,5,7
14	Templates for different drilling patterns used for shot firing.	5
15	Single Shot and multi shot exploder.	7,8,10
16	Smart Classroom fitted with computer and DLP	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Explosives And Blasting Accessories	CO1	12	2	6	6	14
2	II	Blasting Practices in U/G Mines	CO2	12	2	6	6	14
3	III	Safety in Blasting	CO3	10	4	4	6	14
4	IV	Conventional Roof Support	CO4	10	2	4	4	10
5	V	Roof Bolting, Cable Bolting and Yielding Prop	CO5	16	4	6	8	18
Grand Total				**60**	**14**	**26**	**30**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Assignment,Self-Learning and Team Work
- Mid Term Test
- Rubrics for COs

- Seminar/Presentation

Summative Assessment (Assessment of Learning)

- Lab Performance
- End Of Term Examination
- Viva-voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	1	2	3	1	3	3			
CO2	3	2	2	3	1	3	3			
CO3	3	3	2	3	1	3	3			
CO4	3	1	3	3	1	3	3			
CO5	3	1	2	3	1	2	3			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	G.K. Pradhan	Explosive and Blasting Technology	Lovely Prakashan, Dhanbad,ISBN-978-9058096050
2	L.C.Kaku	Mining Digest	Lovely Prakashan, Dhanbad,ISBN Number-13. 978-8179561515
3	Dr.B.P.Verma	Roof Bolting	Lovely Prakashan, Dhanbad,ISBN-13. 978-8179561652
4	D.J.Deshmukh	Elements of Mining Technology	Lovely Prakashan, Dhanbad,ISBN-818990433-7
5	R.D.Singh	Principles & Practices of Modern Coal Mining	New Age International (P) Limited Publishers,New Delhi,ISBN Number-81-224-0974-1
6	Heartman L.Howard	Introductory Mining Engineering	John Wiley & Sons,New York,ISBN Number-0471-62804-2

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.slideshare.net/search?searchfrom=header&q=explosives	Regarding Mining
2	https://www.youtube.com/results?search_query=mining+technology	For Videos on mining
3	https://drive.google.com/file/d/1c-LHOVpvEoySm_68wbGeyVoEHHtUcl6u/view	Explosive and blasting Related Information
4	https://www.e-education.psu.edu/geog000/node/827	Regarding mining videos and books

Sr.No	Link / Portal	Description
5	https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=8zYwEsyFCoiPyJlPmzHDxg==	Mining Related Information

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students